

ASP3231 Assignment 2

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1.

The ideal diffraction limited solution refers to the maximum theoretical resolution a lens can achieve, assuming there are no lens imperfections. The only resolution arises from the physical diffraction itself. The standard approximation defines the angular resolution as

$$\theta \approx \frac{\lambda}{D}. \quad (1)$$

Given a wavelength of 2.15 microns and a 10.5 metre diameter, $\theta = 2.04762 \cdot 10^{-7}$ radians, or 42.24 milliarcseconds. Given the Keck telescope is only about 30% larger than the ideal limit, this indicates that some atmospheric blurring or other imperfections are present on the telescope.

The Strehl ratio is a percentage from 0 to 1 that defines how 'perfect' the peak intensity of the point spread function is. This refers to how the telescope responds to a single point of light, and it can be affected by atmospheric turbulence and impurities. The Strehl ratio of 0.4 indicates it is 40% of the ideal system (which sits at 1.0), which is still considered very good for ground-based telescopes.

(a)

(Brown 2026)

2.

3.

4. References

Brown, M. (2026). *ASP3231 Lecture Videos*. Accessible via Moodle by staff and students of ASP3231.